

Clinical Trial Result Interpretation

- **Statistical significance vs clinical difference**

 Once the clinical trial is completed, the investigators calculate a **p-value** for the endpoints using the collected study results and statistical tests.

 P-value indicate a statistical significance!

 Statistical significance does not automatically mean **a clinical difference** in the effect between the intervention and control groups.

Interpreting Risks and Numbers-Needed-to-Treat

Group	Event/outcome	
	Yes	No
Intervention (treated group)	a	b
Control	c	d

Measures of association description and formulas

- **Terms to Describe Benefit or Harm of Treatment**

 **Experimental event rate (EER):** the proportion of patients in the treated group in whom an event is observed. Thus, if out of 100 treated patients ($a+b$), the event is observed in 27 (a), the event rate is 0.27 ($a/a+b$).

 **Control event rate (CER):** The proportion of patients in the control group in whom an event is observed. $CER = c/c+d$.

Measures of Association Description and Formulas

Terms to Describe Benefit or Harm of Treatment

- ✍ **Absolute risk:** the risk of having a disease at any point in time (incidence). For example, if the incidence of a disease is 1 in 100,000, then the absolute risk is 0.001%.
- ✍ **Absolute risk reduction (ARR):** the difference in the event rate between control group (CER) and treated group (EER): $ARR = CER - EER$.
- ✍ A higher proportion of subjects taking placebo died ($n = 256$ of 2223 or 11.5%) compared to those taking simvastatin (182 of 2221 or 8.2%).
- ✍ The ARR for death associated with simvastatin in this trial equals 3.3% ($ARR = 11.5\% - 8.2\%$); thus, 3.3% of the subjects receiving simvastatin were spared death compared to placebo

NNT

Number needed to treat (NNT):

- ✍ The inverse of the ARR ($1/\text{absolute risk reduction}$)
- ✍ NNT is the number of patients who need to be treated to prevent one bad outcome
- ✍ In this case, it equals 30 ($\text{NNT} = 1/0.033$), meaning 30 subjects need to be treated for a median of 5.4 years with simvastatin instead of placebo to prevent one case of death.
- ✍ It is important to note the **time period** when reporting NNT.

RR

Relative risk (RR):

- ✍ A ratio of two risks, the risk of the outcome/event in the treated group compared with the risk of the outcome in those not exposed (control group)
- ✍ This value is also referred to as risk ratio.
- ✍ Reported as a percentage,
- ✍ Calculated as $RR = EER/CER$.

RR

- **Relative risk (RR):**

- ✍ An **RR equal to 1** indicates no difference between the intervention and control (i.e., the incidence of the outcome was not increased or decreased with the intervention compared to control).
- ✍ The **RR < 1** signifies the intervention lowered the risk of the outcome compared to the control (i.e., protective effect);
⇒ a lower proportion of the intervention group experienced the outcome compared to the control.
- ✍ An **RR > 1** indicates the intervention increased the risk of the outcome; a greater proportion of the intervention group had the outcome compared to control

RR

- **Relative risk (RR):** example,
 - ✍ an RR of death equal to 0.70 was reported in a clinical trial in which subjects were randomized to either **simvastatin** (n = 2221) or **placebo** (n = 2223).
 - ✍ The RR was calculated by dividing the proportion of the subjects who died taking simvastatin (n = 182) by the proportion of those who died taking placebo (n = 256).
 - ✍ $RR = (182/2221)/(256/2223) = 0.7$
 - ✍ The RR of 0.70 indicates simvastatin lowered the risk of death by almost one-third of the baseline risk compared to placebo

RRR

Relative risk reduction (RRR):

- ✓ The proportional reduction in events between control and treated groups. RRR indicates the relative change in the outcome rate between the intervention and control groups.
 - ✓ Reported as a percentage, it is calculated as $(CER - EER)/CER = CER/CER - EER/CER$ or as $1 - RR$.
 - ✓ $RRR = 1 - 0.70 * 100\% = 30\%$.
- ⇒ The risk of experiencing death was 30% lower by treating these subjects with simvastatin instead of placebo

Harm

Absolute risk increase (ARI):

- ✓ The risk difference in outcome rates between treated and control groups when the treatment harms more patients than the control. ARI is calculated as $EER - CER$.

Number needed to harm (NNH):

- ✓ The inverse of ARI ($1/\text{absolute risk increase}$)
- ✓ NNH is the number of patients who would need to be treated to cause one adverse outcome.

Measures of Association Description and Formulas

Measure of Association	Description	Formula
RR	The ratio of risk of an event occurring in one group compared to another group.	$[A/(A + B)]/[C/(C + D)]$ In other words, (% of intervention group with primary endpoint)/(% of control group with primary endpoint)
RRR	Percent of baseline risk removed. The difference in event rates between two groups, expressed as a proportion of the event rate in the untreated group.	$1 - RR$
ARR	Percentage of subjects treated with the intervention spared the adverse outcome compared with the control.	$[C/(C + D)] - [A/(A + B)]$ In other words, (% of control group with primary endpoint) – (% of intervention group with primary endpoint)

Measures of Association Description and Formulas

Measure of Association	Description	Formula
ARI	Percentage of subjects treated with the intervention who experienced the adverse outcome compared with the control.	$[C/(C + D)] - [A/(A + B)]$ In other words, (% of control group with primary endpoint) – (% of intervention group with primary endpoint). The resulting calculation is typically a negative number.
NNT	Number of subjects needed to be treated to prevent one adverse event. A time course is included that represents the average (or median) duration of follow-up during the trial.	$1/(\text{ARR})$
NNH	Number of subjects needed to be treated to prevent for one adverse event to occur. A time course is included that represents the average (or median) duration of follow-up during the trial. The value is referred to as NNH when the ARR is calculated as a negative number.	$1/(\text{ARR})$

Bibliography/References

- Any material that the author uses in the manuscript should be appropriately cited.
- References included in the manuscript should be recent (e.g., outdated articles should not be used unless the results of the article are pertinent to the manuscript) and complete.
- Authors should refrain from extensively citing only their own work.

Acknowledgments

- Individuals contributing to the clinical trial, but who do not meet the requirements for authorship, can be recognized in this section

Funding

- Readers of **industry-sponsored research** should be cognizant of possible conflicts of interest.
- Conflicts of interest arise because the industry may be prompted to publish articles as a means of making their product appear better for a disease state in relation to the standard of care.
- The study design, result presentation style, data interpretations, and study conclusions should be assessed appropriately to determine if the funding source had any influences on the overall clinical trial.

Ten Major Considerations Checklist

- Use these checklists to effectively determine the quality of these trials, in addition to unique considerations to study designs
 - 1) Power of the study set and met (to estimate adequate sample size that will identify a difference between groups if one truly exists)?
 - 2) Dosage/treatment regimen appropriate?
 - 3) Length of study appropriate to show effect?
 - 4) Inclusion criteria adequate to identify target study population?
 - 5) Exclusion criteria adequate to exclude patients who may be harmed in study?

Ten Major Considerations Checklist

- 6) Blinding present?
- 6) Randomization resulted in similar groups?
- 6) Biostatistical tests appropriate for type of data analyzed?
- 6) Measurement(s) standard/validated/accepted?
- 6) Author's conclusions are supported by the results?